SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.* (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100) **Total Marks: 100 Duration: 3hrs**

13. Design a centralized traffic control model using Cloud RAN principles. (BL6)

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

```
clc;
clear;
close all;

%%
=================================================================
% CENTRALIZED TRAFFIC CONTROL MODEL USING CLOUD RAN PRINCIPLES
```

```
%
============================================================
======

%% 1. System Parameters

num_RRU = 6;               % Number of Remote Radio Units
num_users = 120;           % Total number of users
simulation_time = 60;      % Simulation time in seconds

% RRU capacity in Mbps
RRU_capacity = [100 120 100 150 120 100];

% Cloud BBU processing capacity
Cloud_capacity = 600;      % Mbps

% Fronthaul capacity for each RRU
Fronthaul_capacity = 200;  % Mbps

% Traffic demand range per user
min_traffic = 1;           % Mbps
max_traffic = 8;           % Mbps

%% 2. Generate User Traffic

rng(10);    % For repeatable results

user_traffic = min_traffic + ...
   (max_traffic-min_traffic)*rand(num_users,1);

total_requested_traffic = sum(user_traffic);

fprintf('==================================================\n');
fprintf(' CENTRALIZED CLOUD RAN TRAFFIC CONTROL\n');
fprintf('==================================================\n');

fprintf('Number of RRUs            : %d\n',num_RRU);
```

```
fprintf('Number of Users            : %d\n',num_users);
fprintf('Total Requested Traffic     : %.2f Mbps\n', ...
    total_requested_traffic);

%% 3. User Distribution

user_RRU = randi(num_RRU,num_users,1);

%% 4. Initial Traffic Allocation

RRU_traffic = zeros(num_RRU,1);

for u = 1:num_users

    rru = user_RRU(u);

    RRU_traffic(rru) = ...
        RRU_traffic(rru) + user_traffic(u);

end

%% 5. Centralized Traffic Controller

% The Cloud BBU monitors RRU loads and reallocates traffic.

allocated_traffic = zeros(num_RRU,1);

for r = 1:num_RRU

    if RRU_traffic(r) <= RRU_capacity(r)

        allocated_traffic(r) = RRU_traffic(r);

    else

        % Traffic exceeding RRU capacity
        allocated_traffic(r) = RRU_capacity(r);
```

```
    end

end

%% 6. Calculate Blocked Traffic

blocked_traffic = RRU_traffic - allocated_traffic;

blocked_traffic(blocked_traffic < 0) = 0;

total_blocked = sum(blocked_traffic);

%% 7. Centralized Load Balancing

% Find RRUs with available capacity

available_capacity = RRU_capacity' - allocated_traffic;

for r = 1:num_RRU

    if blocked_traffic(r) > 0

        required = blocked_traffic(r);

        % Find RRU with maximum available capacity
        [max_available,target] = max(available_capacity);

        if max_available > 0

            transferred = min(required,max_available);

            allocated_traffic(target) = ...
                allocated_traffic(target) + transferred;

            available_capacity(target) = ...
                available_capacity(target) - transferred;
```

```
            blocked_traffic(r) = ...
                blocked_traffic(r) - transferred;

        end

    end

end

%% 8. Cloud Processing

total_allocated = sum(allocated_traffic);

if total_allocated <= Cloud_capacity

    cloud_status = 'Within Cloud Capacity';

else

    cloud_status = 'Cloud Overloaded';

end

%% 9. Fronthaul Capacity Check

fronthaul_load = allocated_traffic;

fronthaul_overload = ...
    max(0,fronthaul_load-Fronthaul_capacity);

%% 10. RRU Utilization

RRU_utilization = ...
    (allocated_traffic ./ RRU_capacity') * 100;

%% 11. Average Latency Model
```

```
% Simple latency model:
% Base latency + congestion-dependent latency

base_latency = 5;      % ms

latency = base_latency + ...
   0.1 * RRU_utilization;

%% 12. Energy Consumption

% Assume:
% 50 W idle power + 100 W proportional to load

idle_power = 50;
max_dynamic_power = 100;

RRU_power = idle_power + ...
   max_dynamic_power .* (RRU_utilization/100);

total_power = sum(RRU_power);

%% 13. Traffic Efficiency

traffic_efficiency = ...
   (total_allocated / total_requested_traffic)*100;

%% 14. Display RRU Results

fprintf('\n================================================\n');
fprintf(' RRU TRAFFIC STATUS\n');
fprintf('================================================\n');

for r = 1:num_RRU

   fprintf(['RRU %d : Requested = %7.2f Mbps | ' ...
      'Allocated = %7.2f Mbps | Utilization = %6.2f %%\n'], ...
```

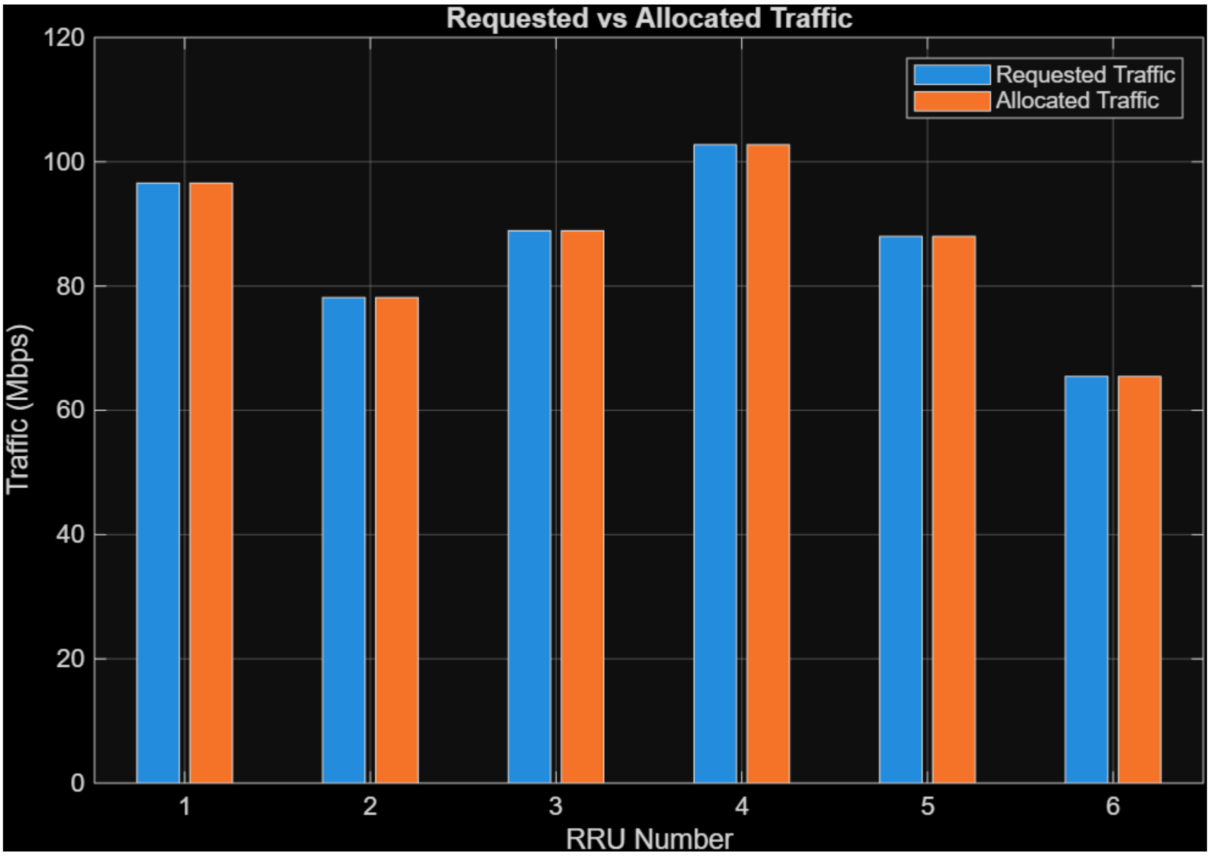
```
        r,RRU_traffic(r),allocated_traffic(r), ...
        RRU_utilization(r));

end

%% 15. Overall Results

fprintf('\n================================================\n');
fprintf(' CENTRALIZED CONTROL RESULTS\n');
fprintf('================================================\n');

fprintf('Total Requested Traffic : %.2f Mbps\n', ...
    total_requested_traffic);

fprintf('Total Allocated Traffic : %.2f Mbps\n', ...
    total_allocated);

fprintf('Total Blocked Traffic   : %.2f Mbps\n', ...
    sum(blocked_traffic));

fprintf('Traffic Efficiency      : %.2f %%\n', ...
    traffic_efficiency);

fprintf('Average RRU Utilization : %.2f %%\n', ...
    mean(RRU_utilization));

fprintf('Average Latency         : %.2f ms\n', ...
    mean(latency));

fprintf('Total Power Consumption : %.2f W\n', ...
    total_power);

fprintf('Cloud Status            : %s\n', ...
    cloud_status);

%% 16. Plot RRU Traffic
```

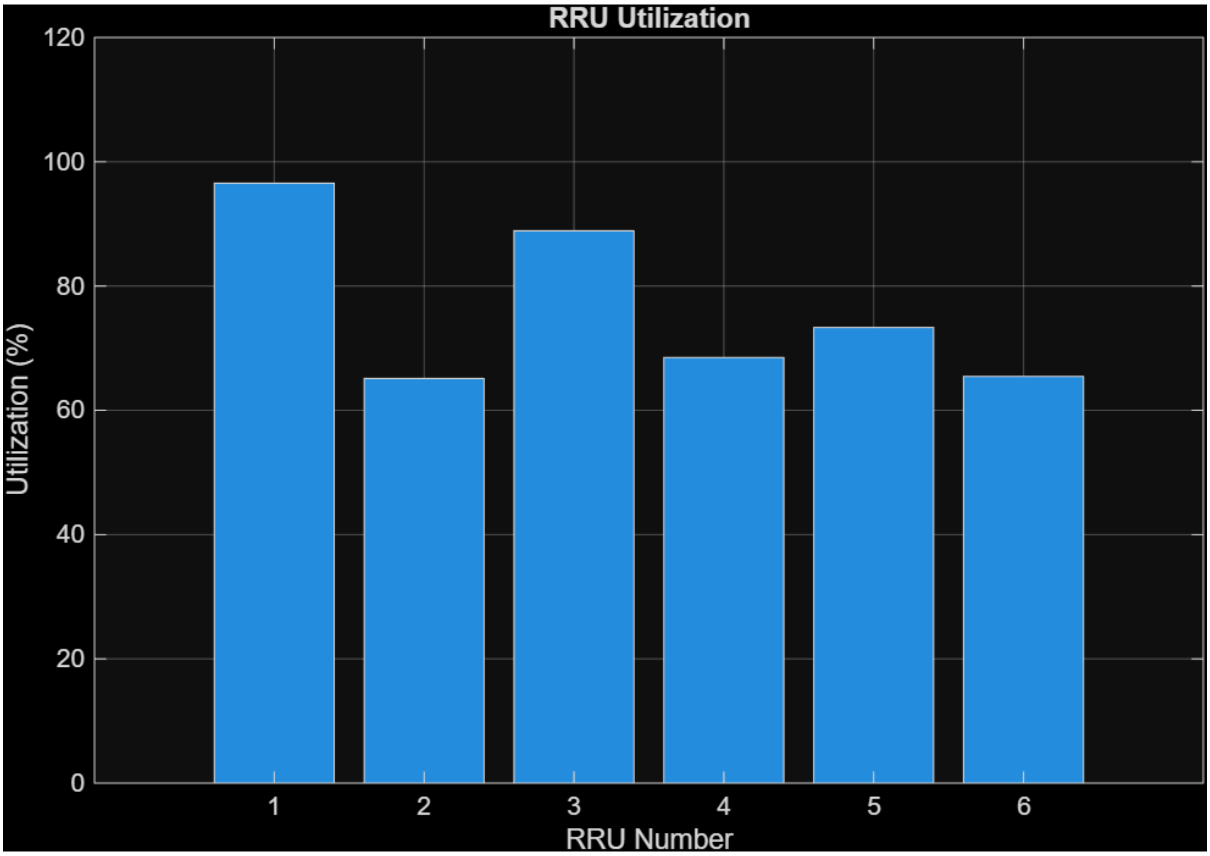
```
figure;

bar(1:num_RRU, ...
   [RRU_traffic allocated_traffic]);

grid on;

xlabel('RRU Number');
ylabel('Traffic (Mbps)');

title('Requested vs Allocated Traffic');

legend('Requested Traffic','Allocated Traffic');

%% 17. RRU Utilization

figure;

bar(1:num_RRU,RRU_utilization);

grid on;

xlabel('RRU Number');
ylabel('Utilization (%)');

title('RRU Utilization');

ylim([0 120]);

%% 18. Latency

figure;

plot(1:num_RRU,latency,'o-','LineWidth',2);

grid on;
```

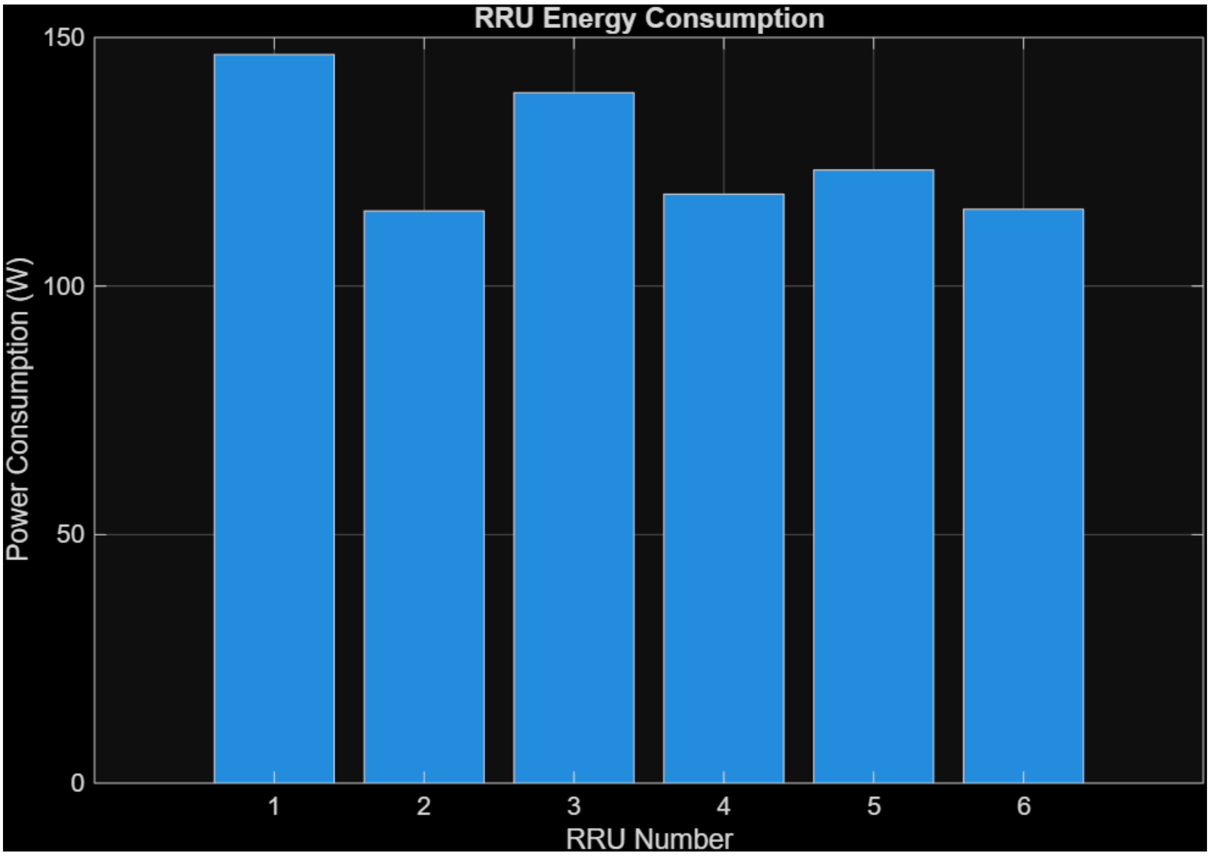
```
xlabel('RRU Number');
ylabel('Latency (ms)');

title('RRU Latency');

%% 19. Energy Consumption

figure;

bar(1:num_RRU,RRU_power);

grid on;

xlabel('RRU Number');
ylabel('Power Consumption (W)');

title('RRU Energy Consumption');

%% 20. Cloud RAN Architecture Visualization

figure;

axis([0 10 0 10]);
axis off;
hold on;

% Cloud BBU
rectangle('Position',[4 7 2 1.2], ...
   'FaceColor',[0.8 0.9 1]);

text(5,7.6,'Cloud BBU','HorizontalAlignment','center');

% RRUs
for r = 1:num_RRU

   x = 1 + mod(r-1,3)*4;
   y = 4 - floor((r-1)/3)*2;
```

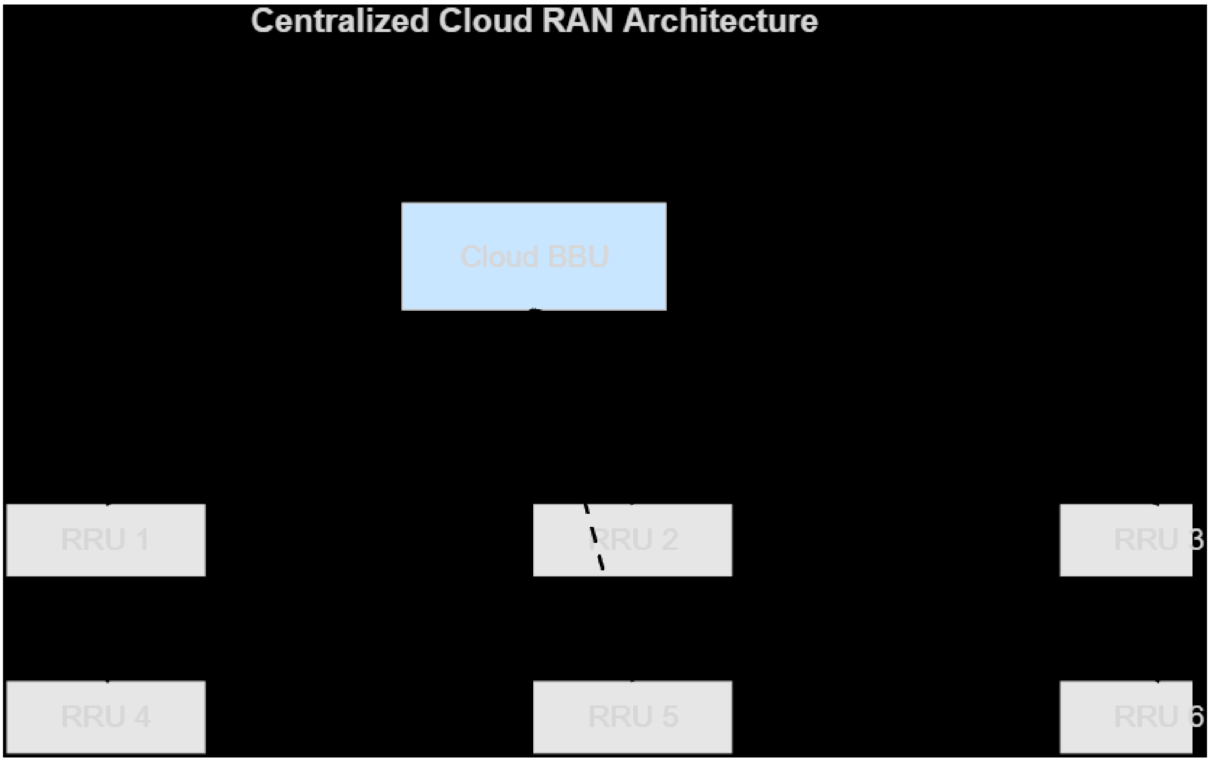
```
    rectangle('Position',[x y 1.5 0.8], ...
        'FaceColor',[0.9 0.9 0.9]);

    text(x+0.75,y+0.4, ...
        sprintf('RRU %d',r), ...
        'HorizontalAlignment','center');

    % Fronthaul connection
    plot([x+0.75 5],[y+0.8 7], ...
        'k--','LineWidth',1.2);

end

title('Centralized Cloud RAN Architecture');

%% 21. Time-Based Traffic Simulation

time = 0:simulation_time;

traffic_time = zeros(length(time),1);

for t = 1:length(time)

    % Random traffic variation
    variation = 0.8 + 0.4*rand;

    traffic_time(t) = ...
        total_requested_traffic * variation;

end

%% 22. Plot Traffic Variation

figure;

plot(time,traffic_time,'LineWidth',2);
```

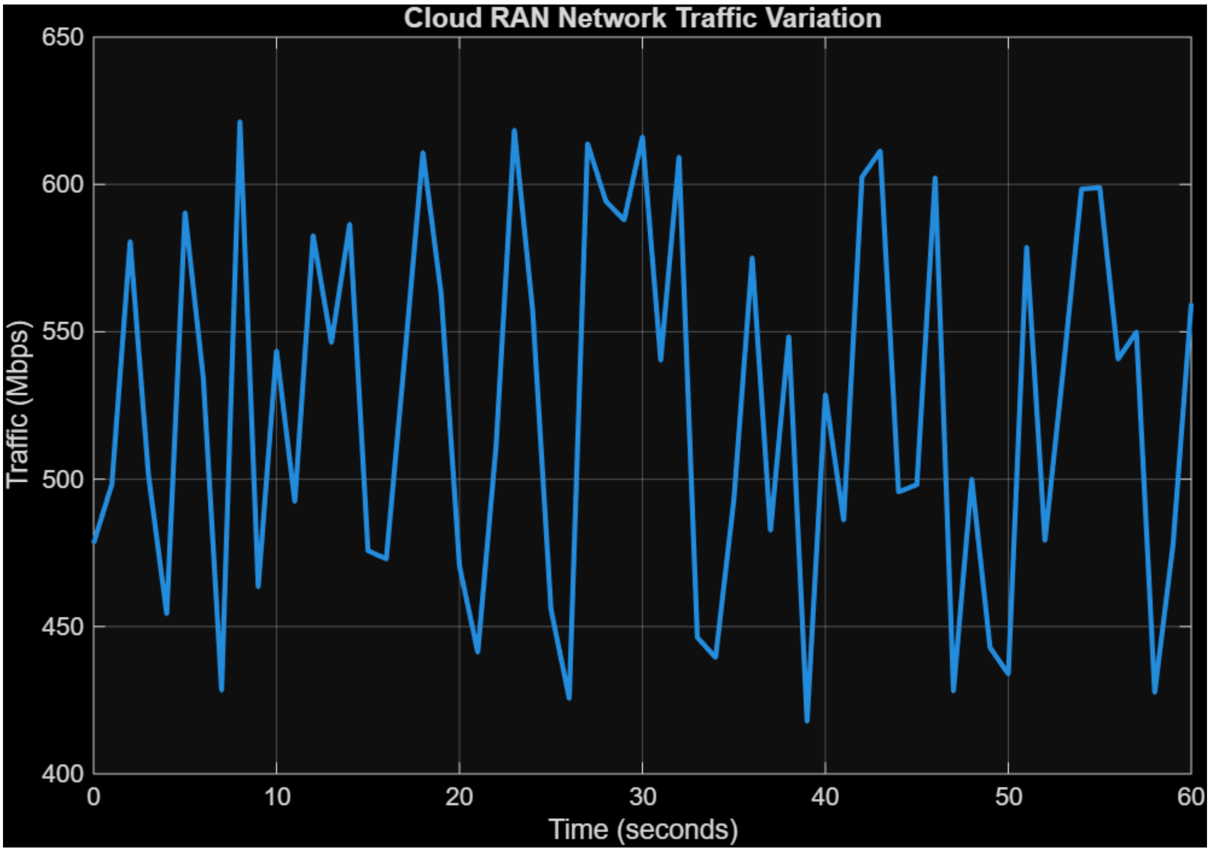
```
grid on;

xlabel('Time (seconds)');
ylabel('Traffic (Mbps)');

title('Cloud RAN Network Traffic Variation');

%% 23. Final Message

fprintf('\n=============================================\n');
fprintf(' Simulation Completed Successfully\n');
fprintf('=============================================\n');
```


Requested vs Allocated Traffic
Requested Traffic
Allocated Traffic
Traffic (Mbps)
RRU Number


RRU Utilization
Utilization (%)
RRU Number


RRU Energy Consumption
150
100
50
0
Power Consumption (W)
1
2
3
4
5
6
RRU Number


Centralized Cloud RAN Architecture
Cloud BBU
RRU 1
RRU 2
RRU 3
RRU 4
RRU 5
RRU 6


Cloud RAN Network Traffic Variation
Traffic (Mbps)
Time (seconds)
650
600
550
500
450
400
0
10
20
30
40
50
60